

W06 - Integration Test Design

ScamGuard Integration / API Testing Package

Team: ScamGuard - แอปตรวจสอบรูปภาพตัดต่อที่ถูกนำมาหลอกลวง

Course: ENGSE601 - Week 06 Output (W06-OUT)

SRS Baseline: 05_Software_Requirement_Specification.md v1.1 (September 12, 2026)

Code Baseline: Panuwat-ta/project branch main, commit `66bc9e4a`

Integration Scope: Mobile/API | Redis | Inference | Source Verification | Heatmap | History/Report

Result: 18 cases - 6 Pass | 11 Fail | 1 Not Ready

Artifact Status: Ready for Review

Date: 2026-09-22

1. Interface Scope and Preparation (W06-01)

Field	Team record
Project/team	ScamGuard - Scam Image Detection Application
Requirement source/version	https://github.com/Panuwat-ta/project/tree/main/Document/srs , branch main, SRS v1.1 dated 2026-09-12
Source-code baseline	https://github.com/Panuwat-ta/project , branch main, commit 66bc9e4a
SRS commit	2b8a1fb0
Execution date	2026-09-22
Primary author / verification owner	ภานุวัฒน์
Assigned peer reviewer	เอกพันธ์ - reviewer decision/sign-off pending
Course activity	Workshop 3: Interface/API Risk Mapping

Interface map

ID	Producer / caller	Consumer / dependency	Contract under test	Result snapshot
I01	Mobile Scan feature	FastAPI Scan API	upload status/id, polling, ownership, cancel	IT-02/03 Pass; IT-01/04 Fail
I02	ScanService	Redis cache	hit/miss, TTL, media consistency	IT-05 Pass; IT-06/07 Fail
I03	ScanService	Inference / ONNX / OCR	mapping, failure propagation, visual contract	IT-08/09 Pass; IT-10 Fail
I04	Scan pipeline	Source Verification	URLs/score/fallback	IT-11 Not Ready; IT-12 Fail
I05	Inference/ScanService	Heatmap storage / API / Mobile	file + /uploads media contract	IT-13 Fail; IT-14 Pass
I06	Completed scan	History / Report	response schema, filters, report contracts	IT-15/16/17/18 Fail

Verification techniques used

- FastAPI ASGI route integration without opening a network port
- deterministic fake DB/Redis and inference stub at explicit service seams
- temporary filesystem for media lifecycle checks
- source/signature contract assertions for interfaces that are visible in code
- **Not Ready** when a required production adapter does not exist instead of inventing a mock production contract

2. Integration Test Design and Results (W06-02)

2.1 Mobile <-> Scan API

ID	Trace	Scenario	Expected	Actual evidence	Result
IT-01	FR-SCAN-02 AC-1	POST multipart JPG	HTTP 202 + identifier	HTTP 200 + id	Fail
IT-02	Polling contract	GET scan in progress	id/status/progress available	200; status=processing_visual, progress=50	Pass
IT-03	Ownership	User B GET User A scan	HTTP 403	HTTP 403	Pass
IT-04	Mobile consumer	DELETE /scan/{id}	cancel contract exists or consumer disabled	HTTP 405 Method Not Allowed	Fail

2.2 ScanService <-> Redis

ID	Trace	Scenario	Expected	Actual evidence	Result
IT-05	FR-SCAN-03 AC-1	cache hit	skip inference and complete	inference call=0; completed	Pass
IT-06	FR-SCAN-03 AC-2	cache miss	write cache with TTL 30 days	JSON serialization fails when empty heatmap_bytes remains bytes; no SETEX	Fail
IT-07	FR-SCAN-03 + FR-XAI-01	cache says heatmap exists but file missing	usable reference or explicit degraded/recovery state	completed with heatmap_image_url=None	Fail

2.3 ScanService <-> Inference / ONNX / OCR

ID	Trace	Scenario	Expected	Actual evidence	Result
IT-08	FR-ANALYSIS-01/02	stub visual=80, AI=0.8, OCR= ตัวน	map result to scan and complete	visual=80; text=25; keyword found; completed	Pass
IT-09	failure boundary	inference raises exception	contain error and mark failed	status= failed , progress=0	Pass
IT-10	FR-ANALYSIS-02 AC-1/2	inspect ONNX worker output contract	separate forgery_confidence and ai_gen_confidence	both fields absent	Fail

2. Integration Test Design and Results (continued)

2.4 Scan Pipeline <-> Source Verification

ID	Trace	Scenario	Expected	Actual evidence	Result
IT-11	FR-ANALYSIS-03 AC-1/2/3	Google Vision returns similar URLs	map URLs/count/context to canonical source score	production Google Vision adapter not present	Not Ready
IT-12	FR-ANALYSIS-03 AC-4	source service down/not connected	source_status="unavailable", URLs empty, source excluded	no source_status; pipeline uses DEFAULT_SOURCE_SCORE	Fail

2.5 Heatmap Storage <-> API <-> Mobile

ID	Trace	Scenario	Expected	Actual evidence	Result
IT-13	FR-XAI-01 AC-1	inference returns heatmap bytes	reference is /uploads/{filename}	reference is absolute filesystem path	Fail
IT-14	FR-XAI-01 AC-1	GET canonical /uploads/heatmaps/w06.jpg	media is served	HTTP 200; bytes match fixture	Pass

2.6 Scan <-> History <-> Report

ID	Trace	Scenario	Expected	Actual evidence	Result
IT-15	FR-HISTORY-01 AC-1	GET history with completed owned scan	body collection scans, item field risk_grade	body uses items, item uses risk_level	Fail
IT-16	FR-HISTORY-01 AC-2/3	date/risk filters	start_date, end_date, risk_grade supported	route has keyword, risk_level; no date range	Fail
IT-17	FR-HISTORY-02 AC-1	create report	HTTP 201, pending, response report_id	HTTP 201/pending, but response uses id	Fail
IT-18	FR-HISTORY-02 AC-2	duplicate report	HTTP 409, no duplicate, canonical detail	HTTP 409/no add, but detail text differs	Fail

Result summary

Result	Count	Meaning
Pass	6	selected integration behavior matches expected contract
Fail	11	current cross-component/API contract does not match baseline
Not Ready	1	required production adapter does not exist for a valid acceptance execution
Total	18	Week 06 integration cases

3. Test Execution and Evidence (W06-03)

Evidence	Type	What it proves
E06-baseline.txt	Baseline record	ENGSE212 commit, SRS version/hash, Python runtime
integration_probe_test.py	Reproducible probe	ASGI/service integration logic and strict canonical assertions
E06-integration-probe.txt	Raw execution	final pytest output: 11 failed, 6 passed in 0.88s
E06-result-register.txt	Result register	per-case actual outcome including IT-11 Not Ready

Execution used an isolated `origin/main` archive. Dummy environment values, fake DB/Redis and deterministic inference are used only at explicit seams. No `.env`, production token, API key, password, production database content or production uploads are included.

The pytest exit code is 1 because canonical assertions intentionally expose 11 implementation/contract failures. It is evidence of product mismatch, not an infrastructure failure.

4. Findings and Disposition (W06-04)

ID	Finding	Status	Owner	Next action
F01	Scan upload returns 200 instead of SRS-required 202	Open	ภาณุวัฒน์	sync route/status contract and re-test
F02	Mobile cancel uses DELETE /scan/{id} but backend has no route	Open	ภาณุวัฒน์	implement cancel or remove/disable consumer action
F03	Cache miss can skip Redis write because empty heatmap_bytes is not JSON serializable	Open	ภาณุวัฒน์	remove/sanitize binary field before json.dumps
F04	Cache can complete with missing heatmap file/reference and no recovery state	Open	ภาณุวัฒน์	define cache/media lifecycle and recovery behavior
F05	Visual worker lacks separate forgery and AI-Gen outputs	Open dependency	ภาณุวัฒน์	fix component contract and re-run I03
F06	Source adapter absent; fallback still injects default source score	Open / Not Ready	ภาณุวัฒน์	implement Google Vision adapter and unavailable fallback
F07	Heatmap response stores filesystem path instead of canonical /uploads URL	Open	ภาณุวัฒน์	normalize server media contract
F08	History response/filter contract differs from SRS	Open	ภาณุวัฒน์	reconcile response/query schema
F09	Report identifier/message contract differs from SRS	Open	ภาณุวัฒน์	synchronize SRS/code response contract

5. Handoff and Review Gate

Destination	Week 06 handoff
Week 07 System/UAT	F01/F02/F05/F06/F07/F08/F09 affect end-to-end user journeys and must not be hidden with mocks
Week 10 Test Data	cache/media/source scenarios need deterministic fixtures and traceable data
Week 11 NFR	real Redis/PostgreSQL, GPU latency and model metrics require dedicated environment
Week 14 Defect Management	unresolved F01-F09 should become formal defect records with lifecycle evidence

Author-side exit status

- Interface map complete: Yes
- 18 cases traceable: Yes
- Raw reproducible evidence: Yes
- Fail/Not Ready preserved: Yes
- Findings have owner/next action: Yes
- AI Use Declaration complete: Yes
- PDF generated and checked: Yes
- Independent human peer review: Pending

Therefore Week 06 is **Ready for Review**, not Submitted. Reviewer decision/sign-off remains blank until a real independent review occurs.

6. AI Use Declaration

รายการ	บันทึก
Week / artifact	Week 06 / Integration Test Design v1
ใช้ AI หรือไม่	Yes

เครื่องมือ/รุ่นเก่าที่ทราบ	ChatGPT GPT-5.6 Sol + Remote Desktop Commander
ใช้เพื่อ	อ่าน SRS/source/handoff, ทำ risk map, สร้างและ execute integration probe, สรุป findings และจัดทำเอกสาร
Requirement input	Canonical Document / srs v1.1
Code input	ENGSE212 main commit 66bc9e4a
การปกป้องข้อมูล	ไม่อ่าน/คัดลอก .env , token, API key, password หรือ production data ลง evidence
Accepted	6 interface groups, 18 cases, explicit test doubles, 11 Fail + 1 Not Ready ตามจริง
Modified	เพิ่ม strict history/report contract assertions หลัง re-check SRS; Google Vision positive case คง Not Ready
Rejected	การสร้าง production adapter สมมติ, การเปลี่ยน Fail เป็น Pass, การสร้าง reviewer sign-off ปลอม
Verification	expected เทียบ SRS v1.1; execute บน isolated baseline; raw output/result register ถูกเก็บ; PDF render ตรวจสอบแล้ว
Verification owner/date	ภาณุวัฒน์ / 2026-09-22

คำยืนยัน: ไม่มีการสร้าง test result, stakeholder decision, approval หรือ peer-review sign-off ที่ไม่ได้เกิดขึ้นจริง